

**GE Drive Systems**

**INSTRUCTIONS**

# 531X175SSBA_M2
# BIU CARD

# Renewal Part

## IMPORTANT
## INFORMATION

**CAUTION:** To ensure proper operation of the motor drive controller, these instructions must be followed for proper set-up and installation of this replacement card.

*These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.*

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# SAFETY PRECAUTIONS

The following paragraphs list some general safety reminders and safety recommendations to be followed when changing the BIU card. These safety precautions will be repeated throughout this instruction book where applicable.

**CAUTION:** Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

**CAUTION:** Power to rack must be turned off before installing or removing any printed circuit card.

**NOTE:** *Read all WARNINGS, CAUTIONS and NOTES in the pertinent instruction book (GEK) prior to removing or installing any printed circuit card.*

## EXTRACTION/INSERTION TOOL

The printed circuit card extraction/insertion tool (Catalog No. IC600MA504A) included with the Series Six Programmable Controller CPU should always be used when installing or removing a card. The BIU card may be located in any addressable I/O slot of a Series Six Programmable Controller high-capacity I/O rack, or Model 60 CPU rack. When located in a CPU rack, the card requires an insertion force of about 50 lbs. (22.68 Kg.). When located in an I/O rack, the card requires about 25 lbs. (11.34 Kg.) insertion force. Use of the extraction/insertion tool should alleviate any problems of possible card damage which could be caused by hand insertion or removal. (See Figure 1.)

## CARD REMOVAL

**NOTE:** *If present, do not disconnect the faceplate grounding wire (connected to inside bottom of faceplate). The purpose of this wire is to ground the faceplate.*

**CAUTION:** Power to rack must be turned off before installing or removing any printed circuit card.

1. Remove the faceplate by turning the thumbscrew one quarter-turn counterclockwise while pulling slightly on faceplate.

2. Insert the card puller studs into the printed circuit card from the soldered side of the card and check that card puller surface is flat against the printed circuit card. (See Figure 2.)

3. Grasp the handle area with either hand and squeeze it to break the card loose from the connectors.

**CAUTION:** Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

4. Remove the card puller and slide the card out of it's slot.

## BIU - SERIES SIX PROGRAMMABLE CONTROLLER BACKPLANE DIP SWITCH SETTINGS

Verify that the seven position DIP switch (physically located on rack backplane, adjacent to each slot) is set as shown in TABLE 1 before installing BIU card in the respective slot. (See Figure 3.)

## BIU CARD JUMPER SETTINGS

This card is a functional replacement for the cards specified in TABLE 2. In order to incorporate design enhancements while continuing to provide compatibility with a variety of previous designs, changes in card layout (including the relocation, addition or deletion of certain jumpers) may have been required. TABLE 2 provides a card number/jumper matrix for cross reference and TABLE 3 shows standard jumper settings. (See Figure 4 for jumper location.) Configure the jumpers of the new card as follows:

1. Locate the card numbers for the old and new cards and refer to TABLE 2.

**NOTE:** *The part number of the card is printed on a white label affixed to the card and begins with the number "531X---". This in NOT the number "F531X---" that is silkscreened directly on the circuit card.*

2. Set jumpers listed as provided on both cards in the same position on the replacement card as they were on the card being replaced. Other jumpers should be set as shown in TABLE 3.

**NOTE:** *If replacement card has expanded memory EEPROMs (U35 and U37), JP19 and JP21 will need to be set in different position than on card being replaced if that card had the lesser EEPROM memory (see TABLE 3).*

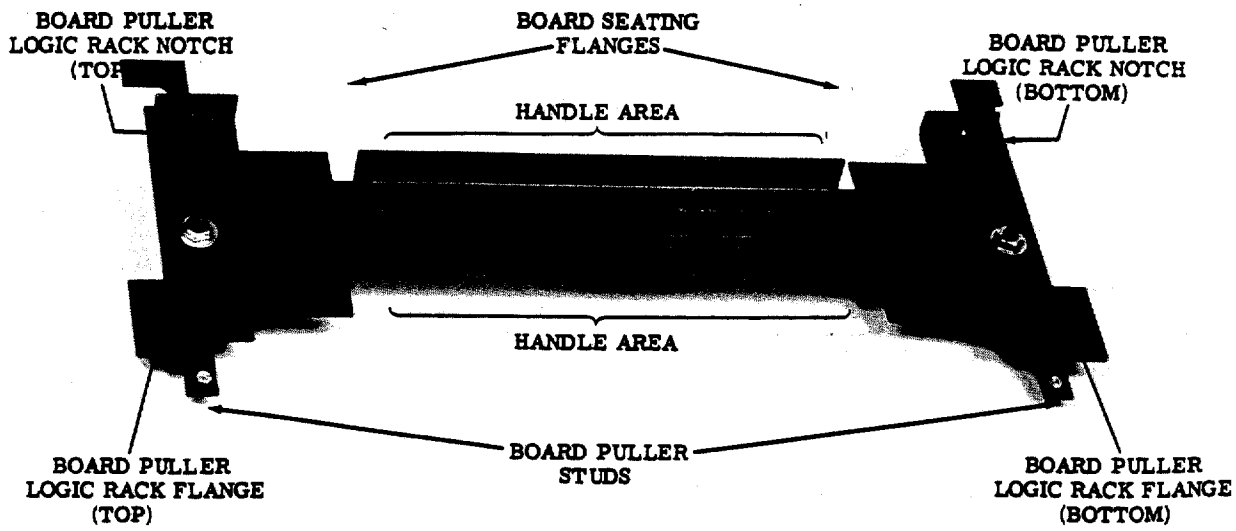



**FIGURE 1. Extraction/Insertion Tool**




**FIGURE 2. Tool Positioned For Card Removal**

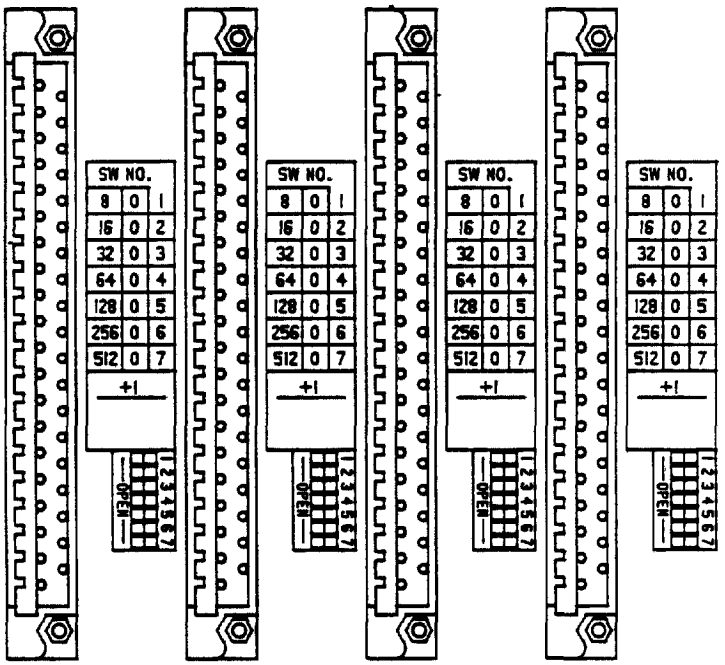



**FIGURE 3. Series Six Backplane DIP Switches**

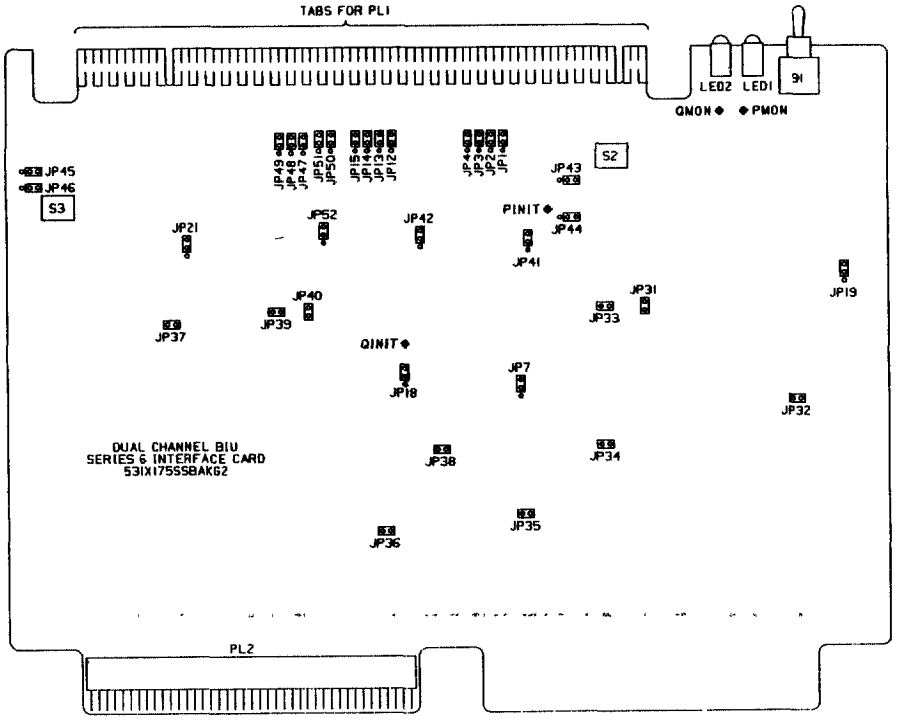



**FIGURE 4. BIU Card Layout**

# TABLE 1: BIU - SERIES SIX PROGRAMMABLE CONTROLLER BACKPLANE DIP SWITCH SETTINGS

| DPREQ Register Content CHNL P | CHNL Q | Switch Position 7654321 | DPREQ Register Content CHNL P | CHNL Q | Switch Position 7654321 |
|---|---|---|---|---|---|
| 1001 | 1009 | ------x | 1497 | 1505 | -xxxxxx |
| 1017 | 1025 | -----xx | 1513 | 1521 | x-----x |
| 1033 | 1041 | ----x-x | 1529 | 1537 | x----xx |
| 1049 | 1057 | ----xxx | 1545 | 1553 | x---x-x |
| 1065 | 1073 | ---x--x | 1561 | 1569 | x---xxx |
| 1081 | 1089 | ---x-xx | 1577 | 1585 | x--x--x |
| 1097 | 1105 | ---xx-x | 1593 | 1601 | x--x-xx |
| 1113 | 1121 | ---xxxx | 1609 | 1617 | x--xx-x |
| 1129 | 1137 | --x---x | 1625 | 1633 | x--xxxx |
| 1145 | 1153 | --x--xx | 1641 | 1649 | x-x---x |
| 1161 | 1169 | --x-x-x | 1657 | 1665 | x-x--xx |
| 1177 | 1185 | --x-xxx | 1673 | 1681 | x-x-x-x |
| 1193 | 1209 | --xx--x | 1689 | 1697 | x-x-xxx |
| 1209 | 1217 | --xx-xx | 1705 | 1713 | x-xx--x |
| 1225 | 1233 | --xxx-x | 1721 | 1729 | x-xx-xx |
| 1241 | 1249 | --xxxxx | 1737 | 1745 | x-xxx-x |
| 1257 | 1265 | -x----x | 1753 | 1761 | x-xxxxx |
| 1273 | 1281 | -x---xx | 1769 | 1777 | xx----x |
| 1289 | 1297 | -x--x-x | 1785 | 1793 | xx---xx |
| 1305 | 1313 | -x--xxx | 1801 | 1809 | xx--x-x |
| 1321 | 1329 | -x-x--x | 1817 | 1825 | xx--xxx |
| 1337 | 1345 | -x-x-xx | 1833 | 1841 | xx-x--x |
| 1353 | 1361 | -x-xx-x | 1849 | 1857 | xx-x-xx |
| 1369 | 1377 | -x-xxxx | 1865 | 1873 | xx-xx-x |
| 1385 | 1393 | -xx---x | 1881 | 1889 | xx-xxxx |
| 1401 | 1409 | -xx--xx | 1897 | 1905 | xxx---x |
| 1417 | 1425 | -xx-x-x | 1913 | 1921 | xxx--xx |
| 1433 | 1441 | -xx-xxx | 1929 | 1937 | xxx-x-x |
| 1449 | 1457 | -xxx--x | 1945 | 1953 | xxx-xxx |
| 1465 | 1473 | -xxx-xx | 1961 | 1969 | xxxx--x |
| 1481 | 1489 | -xxxx-x | 1977 | 1985 | xxxx-xx |

**NOTES:**

1. *DPREQ Register contents are expressed in decimal notation.*
2. *x = Switch in OPEN position (depressed to the left).*
3. *Switch #1 should be in the OPEN position for all conditions.*
4. *Selection of a P & Q channel address actually overlays the selected physical address plus the next 7 locations. (i.e., switch setting -----xx assigns I/O addresses 17 through 24 to P channel and 25 through 32 to Q channel). Care should be taken to not assign other physical uses of these addresses which would cause Series Six Programmable Controller backplane contention problems.*

# TABLE 2: 531X175SSBA_M2 JUMPER MATRIX

| Card Number | Jumpers JP1 | JP2 | JP3 | JP4 | JP7 | JP12 | JP13 | JP14 | JP15 | JP18 |
|---|---|---|---|---|---|---|---|---|---|---|
| AAM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ABM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ACM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ADM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AEM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AFM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AGM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AHM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AJM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AHM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AJM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AKM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ALM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AMM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ANM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| APM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ARM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ASM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |

| Card Number | Jumpers JP19 | JP21 | JP31 | JP32 | JP33 | JP34 | JP35 | JP36 | JP37 | JP38 |
|---|---|---|---|---|---|---|---|---|---|---|
| AAM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ABM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ACM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ADM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AEM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AFM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AGM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AHM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AJM1 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AHM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AJM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AKM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ALM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AMM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ANM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| APM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ARM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ASM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |

**Y = Jumper installed on card.**
**N = Jumper not on card.**

# TABLE 2: 531X175SSBA_M2 JUMPER MATRIX (contd.)

| Card Number | Jumpers JP39 | JP40 | JP41 | JP42 | JP43 | JP44 | JP45 | JP46 | JP47 | JP48 |
|---|---|---|---|---|---|---|---|---|---|---|
| AAM1 | Y | Y | N | N | N | N | N | N | N | N |
| ABM1 | Y | Y | N | N | N | N | N | N | N | N |
| ACM1 | Y | Y | N | N | N | N | N | N | N | N |
| ADM1 | Y | Y | N | N | N | N | N | N | N | N |
| AEM1 | Y | Y | N | N | N | N | N | N | N | N |
| AFM1 | Y | Y | N | N | N | N | N | N | N | N |
| AGM1 | Y | Y | N | N | N | N | N | N | N | N |
| AHM1 | Y | Y | Y | Y | Y | Y | Y | Y | N | N |
| AJM1 | Y | Y | Y | Y | Y | Y | Y | Y | N | N |
| AHM2 | Y | Y | Y | Y | Y | Y | Y | Y | N | N |
| AJM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AKM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ALM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| AMM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ANM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| APM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ARM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| ASM2 | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |

| Card Number | Jumpers JP49 | JP50 | JP51 | JP52 |
|---|---|---|---|---|
| AAM1 | N | N | N | N |
| ABM1 | N | N | N | N |
| ACM1 | N | N | N | N |
| ADM1 | N | N | N | N |
| AEM1 | N | N | N | N |
| AFM1 | N | N | N | N |
| AGM1 | N | N | N | N |
| AHM1 | N | N | N | N |
| AJM1 | N | N | N | N |
| AHM2 | N | N | N | N |
| AJM2 | Y | Y | Y | Y |
| AKM2 | Y | Y | Y | Y |
| ALM2 | Y | Y | Y | Y |
| AMM2 | Y | Y | Y | Y |
| ANM2 | Y | Y | Y | Y |
| APM2 | Y | Y | Y | Y |
| ARM2 | Y | Y | Y | Y |
| ASM2 | Y | Y | Y | Y |

**Y = Jumper installed on card.**
**N = Jumper not on card.**

# TABLE 3: 531X175SSBA_M2 BIU CARD JUMPER POSITIONS

| JUMPER | POSITION | FUNCTION | |
|---|---|---|---|
| JP1 | 1-2 | Test Only (Always in place) | |
| | 2-3 | Not Used | |
| JP2 | 1-2 | Test Only (Always in place) | |
| | 2-3 | Not Used | |
| JP3 | 1-2* | Termination Resistor In | (P Channel) |
| | 2-3 | Termination Resistor Out | (P Channel) |
| JP4 | 1-2* | Termination Resistor In | (P Channel) |
| | 2-3 | Termination Resistor Out | (P Channel) |
| JP7 | 1-2 | Test Only (Always in place) | |
| | 2-3 | Not Used | |
| JP12 | 1-2 | Test Only (Always in place) | |
| | 2-3 | Not Used | |
| JP13 | 1-2 | Test Only (Always in place) | |
| | 2-3 | Not Used | |
| JP14 | 1-2* | Termination Resistor In | (Q Channel) |
| | 2-3 | Termination Resistor Out | (Q Channel) |
| JP15 | 1-2* | Termination Resistor In | (Q Channel) |
| | 2-3 | Termination Resistor Out | (Q Channel) |
| JP18 | 1-2 | Test Only (Always in place) | |
| | 2-3 | Not Used | |
| JP19 | 1-2 | 128K EEPROM (U35) | (P Channel) |
| | 2-3 | 256K EEPROM (U35)** | (P Channel) |
| JP21 | 1-2 | 128K EEPROM (U37) | (Q Channel) |
| | 2-3 | 256K EEPROM (U37)** | (Q Channel) |
| JP31 | 1-2 | Test Only (Always in place) | |
| | — | | |
| JP32 | 1-2 | Test Only (Always in place) | |
| | — | | |

**Termination resistors must be selected if Series Six Programmable Controller is either the first or last connected device on the serial link wires. Both jumpers for a given channel must be set to enable the termination resistor.*

***All 531X175SSBA_M2 BIU cards are furnished with 256K EEPROMs (U35 and U37). These cards are 100% backward compatible with 531X175SSBA_M1 cards that were furnished with 128K EEPROMs (U35 and U37).*

# TABLE 3: 531X175SSBA_M2 BIU CARD JUMPER POSITIONS (Contd.)

| JUMPER | POSITION | FUNCTION | |
|---|---|---|---|
| JP33 | 1-2 | Test Only (Always in place) | |
| | — | | |
| JP34 | 1-2 | Test Only (Always in place) | |
| | — | | |
| JP35 | 1-2 | Test Only (Always in place) | |
| | — | | |
| JP36 | 1-2 | Test Only (Always in place) | |
| | — | | |
| JP37 | 1-2 | Test Only (Always in place) | |
| | — | | |
| JP38 | 1-2 | Test Only (Always in place) | |
| | — | | |
| JP39 | 1-2 | Test Only (Always in place) | |
| | — | | |
| JP40 | 1-2 | Test Only (Always in place) | |
| | — | | |
| JP41 | 1-2 (Def) | Test Only (Always in place) | (Q Channel) |
| | 2-3 | | (Q Channel) |
| JP42 | 1-2 (Def) | Test Only (Always in place) | (P Channel) |
| | 2-3 | | (P Channel) |
| JP43 | 1-2 (Def) | Undefined | (P Channel) |
| | 2-3 | | (P Channel) |
| JP44 | 1-2 (Def) | Undefined | (P Channel) |
| | 2-3 | | (P Channel) |
| JP45 | 1-2 (Def) | Undefined | (Q Channel) |
| | 2-3 | | (Q Channel) |
| JP46 | 1-2 (Def) | Undefined | (Q Channel) |
| | 2-3 | | (Q Channel) |
| JP47 | 1-2 (Def) | Standard LAN | (P Channel) |
| | 2-3 | Optically Isolated LAN*** | (P Channel) |
| JP48 | 1-2 (Def) | Standard LAN | (P Channel) |
| | 2-3 | Optically Isolated LAN*** | (P Channel) |

****The following minimum revision level components are required to utilize the Optically Isolated LAN option:*

| | |
|---|---|
| ***BIU Card*** | *531X175SSBAJM2 or later revision* |
| ***Microapplication Card*** | *531X139APMAAM7 or later revision* |
| ***LAN Power Supply Assembly*** | *36A359133AAG01 or later revision* |

# TABLE 3: 531X175SSBA_M2 BIU CARD JUMPER POSITIONS (Contd.)

| JUMPER | POSITION | FUNCTION | |
|---|---|---|---|
| JP49 | 1-2 (Def) | Standard LAN | (P Channel) |
| | 2-3 | Optically Isolated LAN*** | (P Channel) |
| JP50 | 1-2 (Def) | Standard LAN | (Q Channel) |
| | 2-3 | Optically Isolated LAN*** | (Q Channel) |
| JP51 | 1-2 (Def) | Standard LAN | (Q Channel) |
| | 2-3 | Optically Isolated LAN*** | (Q Channel) |
| JP52 | 1-2 (Def) | Standard LAN | (Q Channel) |
| | 2-3 | Optically Isolated LAN*** | (Q Channel) |

****The following minimum revision level components are required to utilize the Optically Isolated LAN option:*

| | |
|---|---|
| ***BIU Card*** | *531X175SSBAJM2 or later revision* |
| ***Microapplication Card*** | *531X139APMAAM7 or later revision* |
| ***LAN Power Supply Assembly*** | *36A359133AAG01 or later revision* |

## CARD INSERTION

**CAUTION:** Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

**CAUTION:** Power to rack must be turned off before installing or removing any printed circuit card.

1. Orient BIU card with component side of card towards power supply of rack and insert it into the guide.
2. Align the card with the connector on the rack backplane and slide it towards the connector until it has started to seat.
3. Insert the logic rack notch (Top) of the Insertion/Extraction Tool into the short slot beside the top of the solder side of the card. Insert the logic rack notch (Bottom) of the Insertion/Extraction Tool into the short slot beside the botttom of the solder side of the card. (Note that tool is now 180 degrees reversed from removal position; see Figure 6)
4. Grasp the handle area of the tool with either hand and squeeze it until you feel the card seat.
5. Visually inspect the card to be sure it has seated properly, then remove the tool.
6. Reinstall faceplate by placing it in proper position and while pushing in, turn the thumbscrew a quarter-turn clockwise until it feels secure.

NOTE: *LED1 and LED2 on BIU card will blink simultaneously to indicate proper operation when power to Programmable Controller is reapplied. The LEDs will blink alternately when the BIU card begins processing information. If LEDs do not blink as described, recheck card installation per this Instruction Book.*

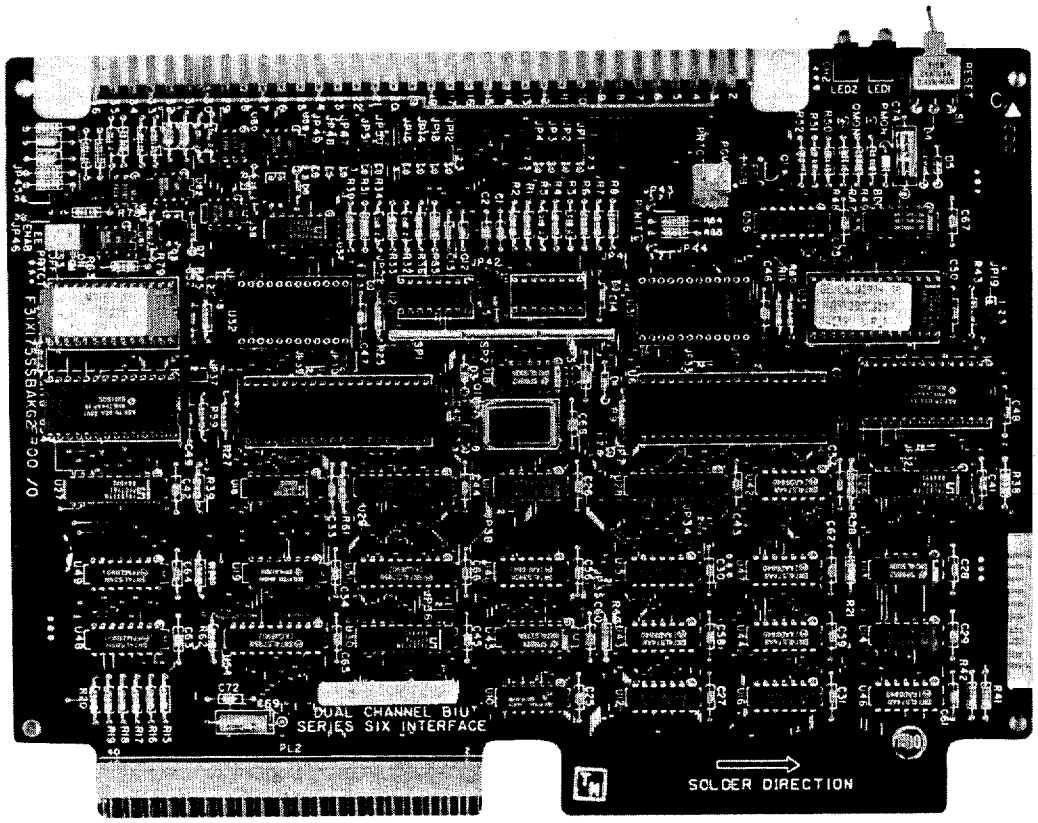



**FIGURE 5. BIU Card**

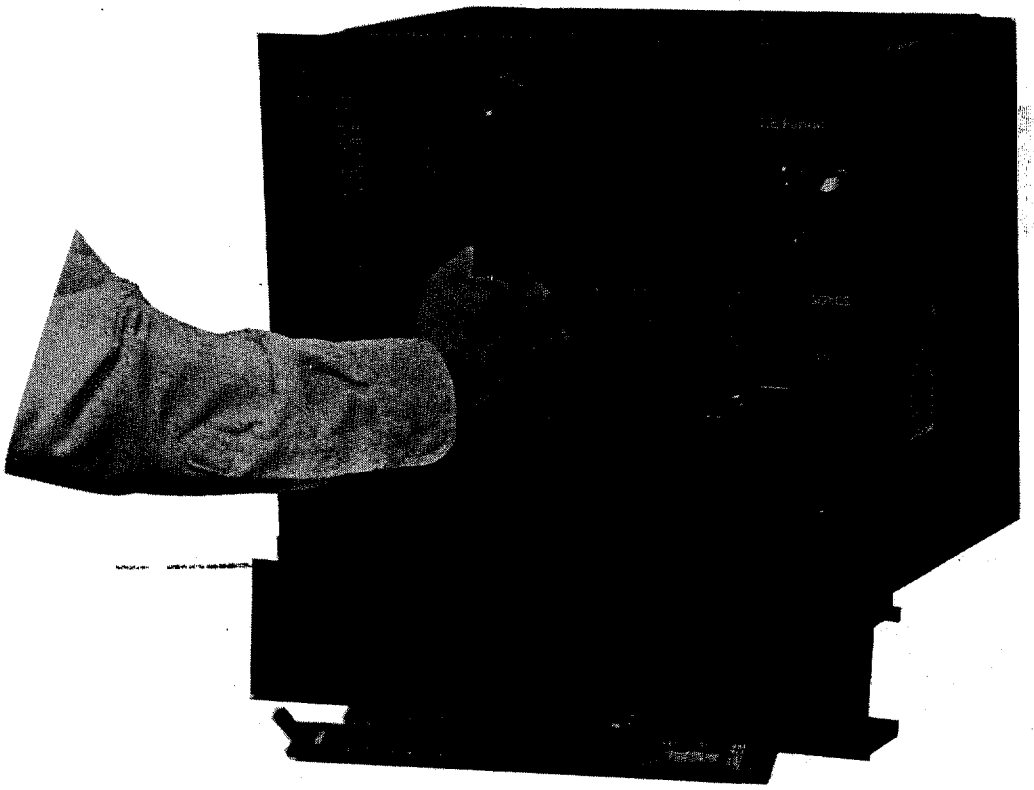


**FIGURE 6. Tool Positioned For Card Insertion**



**GE Drive Systems**

GEK-45151 (1/90)

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*1100 Lawrence Parkway*
*Erie, PA 16531*